Nila

Infant Care Companion

It hears, watches and checks on the baby, on the phone parents already own. Nothing ever leaves it.

On-device AI | 0 network calls | HealthTech

TEAM

Just Us

Nandini K
Team lead ·Full Stack and AI developer

Vishal A S
Computer vision and Ai developer · health tech

INSTITUTIONS

CISAI Centre for Internet Studies & Artificial Intelligence

FOOD HUB

Working Mobile app · Nila 2.9.0 · every AI model runs on the phone

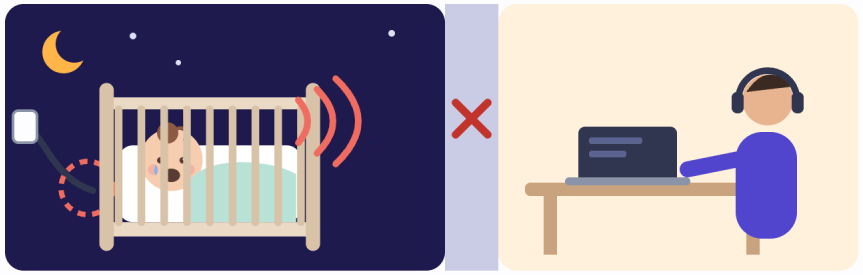
02 · PROBLEM & OPPORTUNITY

Every minute matters. Cloud monitors aren't safe.

THE GAP

Cry unheard

Danger unseen



830,000 **children die of accidents every year, worldwide.** In the US, 947 babies suffocated in bed in 2024.

Drowning · Suffocation in bed · Live wires · Falls, burns, poison

1 in 8
US mothers: postpartum depression

82%
of Americans in nuclear homes

66.5%
US families: both parents work

WHY TODAY'S SMART MONITORS FALL SHORT

8 of 9 graded F
security tests

~55,000
Ring accounts compromised

~13,000
saw others' cameras

$380 + $9.99/mo
and they only notify

HOW TODAY'S TECH CAN HELP

1.26B
smartphones shipped in 2025: the device is already home

45%
of 2026 phones can run generative AI on the device

37%
faster NPU in the Snapdragon 8 Elite Gen 5

0
network calls needed: no cloud to breach

WHO/UNICEF · CDC SUID 2024 · CDC PRAMS 2018 · Pew 2021 · BLS 2024 · Rapid7 2015 · FTC 2023 · Wyze 2024 · Owlet · IDC 2025 · Counterpoint 2026 · Qualcomm

Sense, understand, act, all on the phone

1 · SENSE

- Microphone
- Main + ultra-wide camera
- Home CCTV, found on the Wi-Fi
- Light and motion sensors
- Photos of rooms, skin, medicines

e.g. a cry at 2 a.m.

2 · UNDERSTAND — On-device AI

Audio AI
Cry detection and cause · speech to text

Vision AI
Hazards · safe zone · babyproofing

Language AI
Cited answers · drug interactions

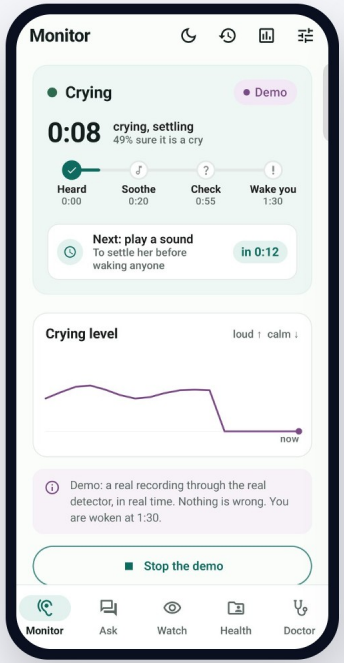
Image processing
Night vision · skin colour · grid tiling

Runs on the phone's NPU, GPU and CPU

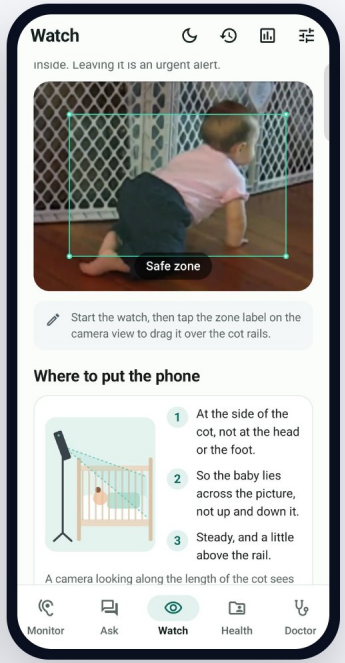
3 · ACT

- Soothes first, then checks it worked
- Distracts with light and a lullaby
- Alerts watch and phone, on home Wi-Fi
- Reminders, records, cited answers

e.g. mum's voice first, then wake her



The cry ladder



The safe zone

Private by architecture: 0 network calls · no account · local network only · no stored passwords · isolated vision AI · source-checked answers

Screens from the Nila app · Android · Kotlin · Qualcomm AI Hub · QNN · ONNX Runtime · LiteRT

Measured on real data, on one phone

BEST RESULT · CRY DETECTOR

96.1% accuracy on babies it never heard

Cries caught	97.3%
AUC	0.992

2,946 test clips · 0.2 ms per check

ON DANGER, IN ~1 S → Light flashes → Lullaby plays → Parent alerted

BABY WATCH · DANGERS IT ALERTS ON

17/18 right · 0 false alarms


Live cord in mouth


Climbing to a window


Reaching a hot pan


Cleaners in reach


Leaning out of a walker


Too close to fire

MORE RESULTS

18/18 **Baby watch**
babies found, safe zone

46/46 **Ask**
right medical sources kept

52/52 **Early signs**
nappy colours, 4 lights

Compared with	Nila	Cloud monitors
Hardware	Your phone	$200–380
Subscription	None	Up to $9.99/mo
Video leaves home	Never	Vendor cloud
Account to breach	None	Yes

WHY IT SCALES

- **$0 server cost** per family
- **One big model** on the NPU
- **Quiet room:** no model runs
- **$1.56B → $3.84B** market by 2034

Held-out test sets on a Snapdragon 8 Elite Gen 5 phone · danger: 18 real baby photos · Fortune Business Insights · Owlet list price

05 · IMPLEMENTATION & SCALING

From a working app to every new parent’s phone

NOW

Working app

Nila 2.9.0 on a Snapdragon 8 Elite Gen 5 · 522 tests pass

NEXT 3 MONTHS

Validate

Clinical study with paediatricians · Play Store beta

6–12 MONTHS

Expand

More phones and NPUs · iOS · more languages

12–24 MONTHS

Distribute

Phone-maker pre-install · hospital programmes

CHALLENGES, AND HOW WE HANDLE THEM

Challenge	Our approach
Big models, a hot phone	One model on the NPU; looks only on events
Many phones and chips	Three backends, smaller fallback models
False alarms lose trust	Guard rails: 11/18 → 17/18, 0 false alarms
Updates with no cloud	Signed model packs; no user data returns
Regulation	An aid: never diagnoses or states a dose

THE SAME ENGINE SCALES TO

- **Elder care**
- **Hospital rooms**
- **Home security**
- **Childcare centres**

☆ Learns on the phone what settles *each* baby